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ScienceDirect
Advanced Engineering Informatics 00 (2022) 1–16
Advanced Engineering Informatics

Highly Imbalanced Fault Diagnosis of Gas Turbines Using Clustering-based Downsampling and Deep Siamese Self-Attention Network

Abstract

For highly reliable gas turbines that rarely suffer faults, the overwhelming majority of historical data are collected under healthy state, while only a very small number of them are fault samples. However, traditional deep neural networks pay most attention to normal samples, resulting in a high missing diagnostic rate for fault samples. To address this problem, this paper develops a new fault diagnosis framework that integrates clustering-based downsampling with deep Siamese self-attention network (CBU-DSSAN), to reduce the number of normal training samples via clustering and strengthen the ability of fault feature extraction via multi-head self-attention mechanism. First, clustering-based downsampling is only conducted on the normal samples, and the cluster centers are put together with the fault samples to serve as the training dataset, in order to balance the normal and fault classes. Second, the Siamese network maps the original data set into an embedded feature space, in which fault samples and normal samples belonging to different classes are far away from each other. Finally, the performance of the developed CBU-DSSAN has been evaluated using actual monitoring data of gas turbines.

Keywords:
Gas turbines, Imbalanced fault diagnosis, Clustering-based downsampling, Deep Siamese self-attention network, Multi-head self-attention mechanism

1. Introduction

As a core component of airplanes, the reliability of gas turbines has a decisive influence on flight safety. However, due to the long-time high-temperature, high-pressure, and high-speed operating environment, it is inevitable for gas turbines to encounter performance degradation and even faults, such as exhaust gas temperature indication faults (EIF) and total air temperature sensor indication faults (TSIF) [1][2]. If the early minor fault cannot be diagnosed in time, it may gradually evolve into a catastrophic failure, which will lead to severe disasters and high maintenance costs [3][4]. Therefore, to ensure flight safety and reduce economic losses, it is necessary to carry out real-time performance monitoring and fault diagnosis.

The fault diagnosis methods for gas turbines are usually categorized into two classes, including physics-based methods and data-driven methods. The physics-based method requires establishing physical performance baseline models for gas turbine fault diagnosis, and the diagnosis results can be explained directly [5][6]. For example, Kobayashi et al.[5] developed a piecewise linear model to approximate the transient nonlinear response of the engine for fault detection. Tan et al.[6] proposed a residual useful life prediction model of engine turbine blades using trans-layer model learning. However, their limitations in engineering practice are also obvious. First, a large amount of prior knowledge and even some unmeasurable parameters are required in the modeling process. Second, it is a great challenge to establish accurate physical models for gas turbines, because the damage propagation processes and dynamic responses of complex equipment are very complicated [7].

In contrast, data-driven methods learn a classification model from monitoring data and apply the classification model to infer the health status of gas turbines [8]. As an important branch of the data-driven methods, deep learning methods can automatically learn useful and informative features from historical monitoring data [9]. That is, it is not necessary to manually select features, which makes fault diagnosis much more direct [10]. In recent years, some deep learning-based methods have emerged and achieved encouraging performance in the field of fault diagnosis [11]. For example, Zhao et al. [12] developed deep residual shrinkage networks to improve the feature learning ability from highly noised monitoring data, thus improving the fault diagnosis accuracy of rotating machinery. Zhang et al. [13] proposed an adaptive loss-weighted meta-residual network to address the wind turbine gearbox fault diagnosis with noisy labels. Zhao et al. [14] applied an enhanced bi-directional gated recurrent unit network (GRU) to learn local features, and verified the effectiveness and generalizability of the model on tool wear prediction and incipient fault diagnosis of gearboxes and bearings. Yu et al. [15] built a deep belief network using a conditional restricted Boltzmann machines, and accurately predicted the health status of hydraulic pumps. Therefore, the effectiveness of the deep learning method in mechanical fault diagnosis has been verified under the condition of sufficient and balanced training samples.

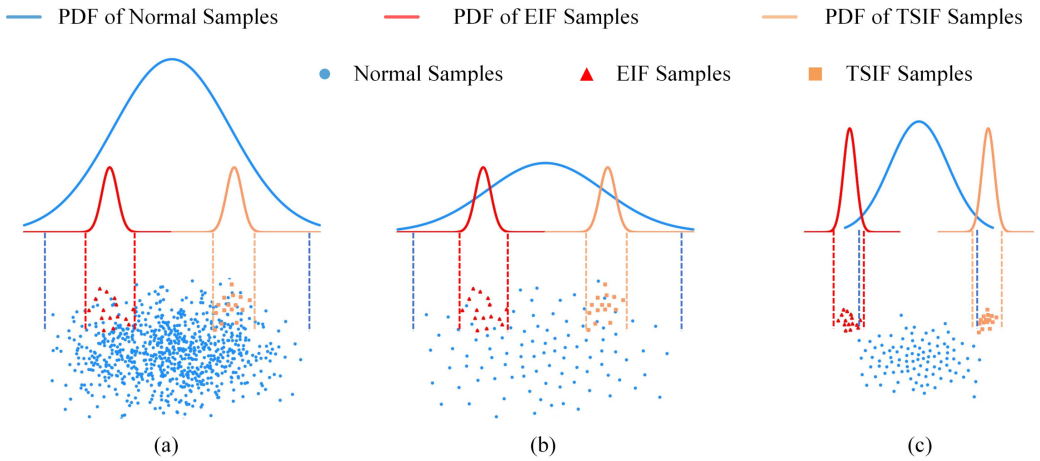


Figure 1. 2D sketch maps of (a) samples in a traditional feature space, (b) normal clustered centers and fault samples after clustering-based downsampling, (c) normal clustered centers and fault samples in the learned feature space using CBU-DSSAN. (PDF represents the probability density function of the sample distribution.)

However, a challenge faced by traditional deep learning methods is that labeled fault samples are very limited, which are far less than normal samples. The reasons are that on one hand, gas turbines, as mature industrial products, rarely fail; on the other hand, gas turbines may suffer many types of faults, and it is difficult to collect sufficient samples under the same fault state [16]. Moreover, another challenge is that fault samples and normal samples are often highly overlapped, as depicted in Fig.1(a). The probability density of normal samples in the overlapping area is likely to be higher than that of fault samples, and traditional classification models have difficulty effectively distinguishing them. In this case, all samples tend to be diagnosed as normal samples, which may lead to an unsatisfactory diagnostic result for fault samples [17]. Therefore, a significant task is to reduce the degree of class imbalance and overlap.

Aiming at the problems, the normal samples can be resampled using downsampling strategies. The sample distribution may become like that depicted in Fig.1(b), which is obviously more desirable than that in Fig.1(a). Because the probability density of fault samples falling within the central part of the overlapping area is higher than that of normal samples, these fault samples can be correctly diagnosed. Notably, traditional downsampling strategies randomly select a subset from normal samples and probably eliminate some representative data [17]. To select the representative data as much as possible, we select the cluster centers of normal samples and then put them together with fault samples to form the new training dataset, which is called clustering-based downsampling approach. Clustering methods, such as k-means, group normal samples into a number of clusters, and each cluster contains similar data [18]. The cluster center is the point closest to all the samples that belong to that cluster; from this point of view, it is the most suitable to represent that cluster.

However, as indicated in Fig.1(b), the probability density of fault samples falling in the border part of the overlapping area is still lower than that of normal samples. To further enhance the separability of samples of different classes, we adopt the deep Siamese self-attention network to map the original data set into an embedded feature space, in which sample pairs from the same classes are close to each other, while sample pairs from the different classes are far away from each other during the training stage [19]. That is, normal samples and fault samples are more separable in the embedded feature space, as depicted in Fig.1(c). Besides, some other benefits of the deep Siamese self-attention network are described below. First, the Siamese network takes sample pairs as its input. The number of training sample pairs is sufficient, which can alleviate the over-fitting problem to some extent. Second, the SANet, which is used as the feature extraction module of the Siamese network, can effectively deal with multi-dimensional time-series data [20].

In this paper, we develop a novel gas turbine fault diagnosis framework that integrates clustering-based downsampling with a deep Siamese self-attention network. Moreover, the main contributions of this paper are as follows:

- The clustering-based downsampling is employed to construct a representative subset of normal samples, and then put them together with fault samples as the training data set, thereby reducing the degree of class imbalance and overlap.
- The Siamese network maps the original data set into an embedded feature space, in which normal and fault samples belonging to different class pairs are more separable.
- The self-attention network, which is used as the feature extraction module of the Siamese network, can extract abundant temporal features from multi-dimensional time-series data.
- The actual gas turbine monitoring data of an Asian airline from 2003 to 2012 are used to evaluate the performance of the proposed model.

The remainder of this article is organized as below. First, the motivations and theory of the CBU-DSSAN are described in Section 2. Then, Section 3 verifies the validity of the proposed model using actual monitoring data of the gas turbine. Finally, Section 4 comes to the main conclusion and the future research work prospects.

2. Motivations and Theory of the developed CBU-DSSAN

2.1. Motivations

For gas turbines, it is extremely important to apply accurate and robust methods to diagnose faults. However, the diagnosis process faces four significant challenges, which are described as follows.

- **Neglection of fault samples** Fault samples are far less than normal samples in the training set. During the training phase, the total gradient is dominated by normal samples for deep neural networks; that is, deep neural networks pay more attention to the training of normal samples [21]. Even if all training samples are diagnosed as normal samples, the cross-entropy loss is still small. In this case, fault samples are easily misclassified as normal samples.
- **Overfitting** Deep neural networks contain a large number of parameters that need to be optimized. If only limited labeled samples are used to train deep neural networks, they easily fall into over-fitting [22]. That is, they perform perfectly on the training set while fitting poorly on the test set.
- **Class overlapping** When gas turbines suffer minor faults, fault samples are close to normal samples, as sketched in Fig.1(a) and Fig.1(b). In this case, classifiers have difficulty effectively distinguishing between different classes, and fault samples belonging to the minority class are easily classified into the majority class [17].
- **Unclear time-series correlations** The monitoring data of gas turbines belong to typical multidimensional time series. There are unclear time-series correlations and dependencies, and how to extract all temporal features from monitoring data is a great challenge [23].

2.2. Key characteristic parameter selection and sample construction

The gas turbine is a highly complex thermal machine composed of the fan, low-pressure compressor (LPC), high-pressure compressor (HPC), combustion chamber (CC), high-pressure turbine (HPT), and low-pressure turbine (LPT), as shown in Fig.2(a). Relevant studies [24][25] show that the state characteristic parameters of gas turbines can reflect their performance status; therefore, it is an essential step for fault diagnosis to select key characteristic parameters. Among many state characteristic parameters, exhaust gas temperature (EGT), low pressure compressor rotor speed (N1), core speed (N2), and fuel flow (FF) play an important role in monitoring gas turbine performance degradation and fault diagnosis [16].

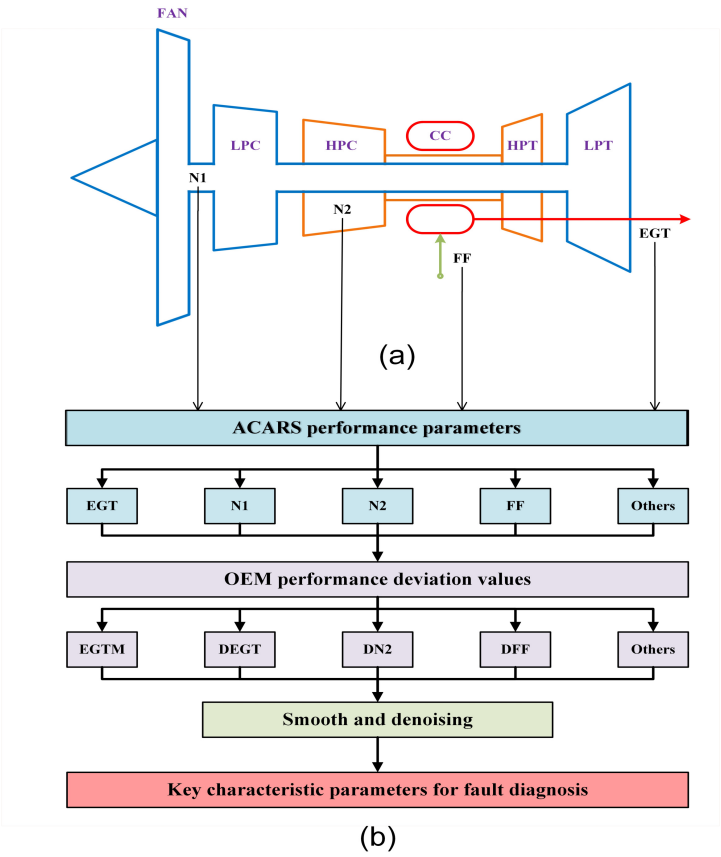


Figure 2. (a) Structure sketch map of the objective gas turbine, (b) Acquisition and preprocessing flow of gas turbine monitoring data

However, in actual flight, the measured values of the above-mentioned characteristic parameters are also affected by environmental conditions, operating conditions, thrust, and so on. Direct use of the measured values may misjudge the performance status of gas turbines. To more accurately reflect the gas turbine performance status, original equipment manufacturer (OEM) converts them into the baseline deviation values according to the baseline model, variable environmental conditions, and thrust. For example, as depicted in Fig.2(b), general electric (GE) company converts the EGT, N1, N2, and FF to exhaust gas temperature margin (EGTM), delta exhaust gas temperature (DEGT), delta fuel flow (DFF), and delta core speed (DN2) according to flight altitude, Mach number, and thrust.

Based on the above analysis, EGTM, DEGT, DN2, and DFF are selected as key characteristic parameters of gas turbine fault diagnosis in this paper. These key characteristic parameters can reflect the performance status of gas turbines. For example, Fig.3 shows the change trend of the key characteristic parameters when the EGT indication fault occurs. DFF and DN2 fluctuate in a small range, but DEGT shows an obvious upward trend.

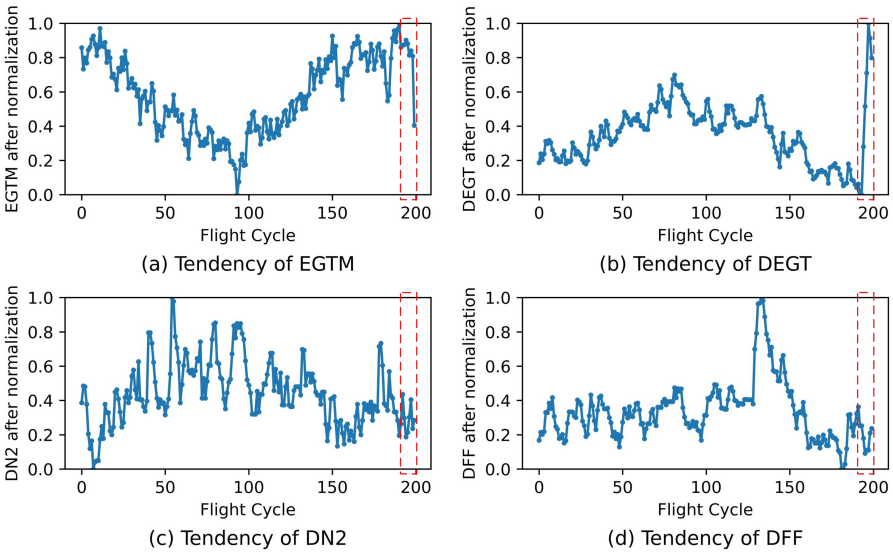


Figure 3. Tendency of key characteristic parameters before a fault occurs in gas turbine (The red dotted rectangular boxes indicate the sliding window with a length of ten.)

In this paper, samples are constructed using a sliding window method. The construction process includes three steps and is described as follows.

First of all, according to the maintenance reports, the removal date of the gas turbine can be found out. Then, the m_k flight-cycle key characteristic parameters before the removal data of the kth gas turbine can be obtained from the historical monitoring data. The key characteristic parameters are as follows:

$$KCP_k = \begin{bmatrix} EGTM_{k,1} & EGTM_{k,2} & \cdots & EGTM_{k,m_k} \\ DEGT_{k,1} & DEGT_{k,2} & \cdots & DEGT_{k,m_k} \\ DN2_{k,1} & DN2_{k,2} & \cdots & DN2_{k,m_k} \\ DFF_{k,1} & DFF_{k,2} & \cdots & DFF_{k,m_k} \end{bmatrix} \tag{1}$$

Second, according to customer notification reports (CNR), it has been found that the length of the fault sequence is between 5 and 120, and the vast majority is not more than 10. The size of the sliding window is finally set to 4×10, and the step length is set to 10, as depicted in Fig.3. The sliding window moves from end (i.e., the flight cycle in which fault occurs) to front. Therefore, the pth fault symptom sample (FSS) of the kth gas turbine is defined as follows:

$$FSS_{k,p} = \begin{bmatrix} EGTM_{k,m_k-10\times p+1} & EGTM_{k,m_k-10\times p+2} & \cdots & EGTM_{k,m_k-10\times p+10} \\ DEGT_{k,m_k-10\times p+1} & DEGT_{k,m_k-10\times p+2} & \cdots & DEGT_{k,m_k-10\times p+10} \\ DN2_{k,m_k-10\times p+1} & DN2_{k,m_k-10\times p+2} & \cdots & DN2_{k,m_k-10\times p+10} \\ DFF_{k,m_k-10\times p+1} & DFF_{k,m_k-10\times p+2} & \cdots & DFF_{k,m_k-10\times p+10} \end{bmatrix} \tag{2}$$

When $p = 1$, $FSS_{k,p}$ represents the fault sample; Otherwise, $FSS_{k,p}$ represents the normal sample, and p is a positive integer.

Finally, for the convenience of description, the set of normal samples is denoted as $D_{NOR} = \{FSS_{NOR_i} : i = 1, 2, \cdots, N_{NOR}\}$. FSS_{NOR_i} represents the ith normal sample and N_{NOR} is the number of normal samples. The set of EGT indication fault samples is denoted as $D_{EIF} = \{FSS_{EIF_i} : i = 1, 2, \cdots, N_{EIF}\}$. FSS_{EIF_i} represented the ith fault sample and N_{EIF} is the number of EGT indication fault samples. The set of TAT sensor indication fault samples is denoted as $D_{TSIF} = \{FSS_{TSIF_i} : i = 1, 2, \cdots, N_{TSIF}\}$. FSS_{TSIF_i} represents the ith fault sample and N_{TSIF} is the number of TAT sensor indication fault samples. To avoid the randomness of one experiment, this paper adopts five-fold cross-validation to divide the data set. The data set is divided into five disjoint subsets. Each time one subset is selected as the test set and the others are the training set. That is, $D_{NOR} = D_{NOR}^{(TRAIN)} \bigcup D_{NOR}^{(TEST)}$, $D_{EIF} = D_{EIF}^{(TRAIN)} \bigcup D_{EIF}^{(TEST)}$

and $D_{TSIF} = D_{TSIF}^{(TRAIN)} \bigcup D_{TSIF}^{(TEST)}$. The training set is $D^{(TRAIN)} = D_{NOR}^{(TRAIN)} \bigcup D_{EIF}^{(TRAIN)} \bigcup D_{TSIF}^{(TRAIN)}$, and the test set is $D^{(TEST)} = D_{NOR}^{(TEST)} \bigcup D_{EIF}^{(TEST)} \bigcup D_{TSIF}^{(TEST)}$.

2.3. Clustering-based downsampling for balancing different classes

To alleviate the imbalanced class problem, an ideal solution is to select a representative subset of normal samples, and then then put them together with fault samples as a balanced training set. This paper adopts clustering algorithms to group normal samples into a suitable number of clusters, and then select cluster centers as the aforementioned representative subset. Although the clustering algorithms may result in information loss (i.e., cluster centers cannot completely represent the original normal samples), there are some positive benefits when using these clustering algorithms. The basic idea of general clustering algorithms is described as below.

Suppose $D_{NOR}^{(TRAIN)} = \left\{FSS_{NOR_i} : i = 1, 2, \cdots, N_{NOR}^{(TRAIN)}\right\}$ is partitioned into $K_{NOR}\left(K_{NOR} < N_{NOR}^{(TRAIN)}\right)$ clusters and the set of the clusters is $CD_{NOR} = \{CD_{NOR_i} : i = 1, 2, \cdots, K_{NOR}\}$, then the clustering objective function is shown as follows:

$$CD_{NOR}^{*} = \underset{CD_{NOR}}{argmin} \sum_{i=1}^{K_{NOR}} \sum_{FSS_{NOR_j} \in CD_{NOR_i}} \left[D_E\left(CCD_{NOR_i}, FSS_{NOR_j}\right)\right]^2 \tag{3}$$

where, CCD_{NOR_i} is the cluster centers of the ith cluster CD_{NOR_i} and $D_E(\cdot,\cdot)$ is the Euclidean distance. In Euclidean space, when $K_{NOR} \geq 2$ and the dimension of $FSS_{NOR_i} \geq 2$, this is an NP-hard optimization problem. That is, it is extremely difficult to find a global optimum set of cluster centers $CCD_{NOR} = \{CCD_{NOR_i} : i = 1, 2, \cdots, K_{NOR}\}$ [26]. The k-means method has the ability to find a relatively optimal set of cluster centers with efficient computation time, which has been applied widely in many fields [27][28]. In this paper, the k-means method is selected to find a set of cluster centers CCD_{NOR} of normal training samples, and the CCD_{NOR} is put together with the fault samples to serve as the balanced training dataset $BD^{(TRAIN)} = CCD_{NOR} \bigcup D_{EIF}^{(TRAIN)} \bigcup D_{TSIF}^{(TRAIN)}$.

2.4. Mapping learning via deep Siamese self-attention network

2.4.1. Construction of sample pairs

After the selection of the training dataset $BD^{(TRAIN)}$, sample pairs for training the Siamese network need to be constructed, denoted as $SD^{(TRAIN)}$, and can be expressed by

$$SD^{(TRAIN)} = \left\{ \left\langle FSS_k, \widehat{FSS_k} \right\rangle : FSS_k, \widehat{FSS_k} \in BD^{(TRAIN)}, k = 1, 2, \cdots, N_{SD}^{(TRAIN)} \right\} \tag{4}$$

where FSS_k and $\widehat{FSS_k}$ represent two samples that are selected from $BD^{(TRAIN)}$. In addition, $\langle\cdot,\cdot\rangle$ represents a sample pair, and $N_{SD}^{(TRAIN)}$ is the number of sample pairs. The setting rule of label y_k of sample pair $\left\langle FSS_k, \widehat{FSS_k} \right\rangle$ is expressed by

$$y_k = \begin{cases} 1 & \left(FSS_k, \widehat{FSS_k} \in CCD_{NOR}\right) or \left(FSS_k, \widehat{FSS_k} \in D_{EIF}^{(TRAIN)}\right) or \left(FSS_k, \widehat{FSS_k} \in D_{TSIF}^{(TRAIN)}\right) \\ 0 & otherwise \end{cases} \tag{5}$$

That is, the label of the sample pair belonging to the same class is 1, and that of the sample pair belonging to different classes is 0.

2.4.2. Siamese network architecture

The Siamese network is one of the most famous few-shot learning methods in machine fault diagnosis [29]. A Siamese network generally contains the feature extraction module and similarity calculation module. The feature extraction module takes the form of sample pair $\left\langle FSS_k, \widehat{FSS_k} \right\rangle$ as input and learn a mapping $F_W(\cdot)$ that maps sample pairs to an embedded feature space. And $\left\langle F_W\left(FSS_k\right), F_W\left(\widehat{FSS_k}\right)\right\rangle$ is the corresponding embedded representation of the sample pair. The similarity calculation module uses a simple distance metric in the feature space, such as Euclidean distance $D_E(\cdot,\cdot)$, to approximate the similarity of the sample pair.

Contrastive loss is usually used to train the Siamese network, effectively dealing with paired data [30]. And, it is defined as below.

$$L_{sia} = \frac{1}{N_{SD}^{(TRAIN)}} \sum_{k=1}^{N_{SD}^{(TRAIN)}} y_k D_E\left(F_W\left(FSS_k\right), F_W\left(\widehat{FSS_k}\right)\right) + (1-y_k)\, max\left[d - D_E\left(F_W\left(FSS_k\right), F_W\left(\widehat{FSS_k}\right)\right), 0\right] \quad (6)$$

where d represents a margin. According to contrastive loss, if a sample pair $\left\langle FSS_k, \widehat{FSS_k}\right\rangle$ belongs to the same class, $D_E\left(F_W\left(FSS_k\right), F_W\left(\widehat{FSS_k}\right)\right)$ will be smaller and smaller during the training stage. That is, $F_W\left(FSS_k\right)$ and $F_W\left(\widehat{FSS_k}\right)$ are closer and closer to each other. If $\left\langle FSS_k, \widehat{FSS_k}\right\rangle$ belongs to the different classes, $D_E\left(F_W\left(FSS_k\right), F_W\left(\widehat{FSS_k}\right)\right)$ will be larger and larger. That is, $F_W\left(FSS_k\right)$ and $F_W\left(\widehat{FSS_k}\right)$ are further and further away from each other. Therefore, normal and fault samples belonging to different classes are more separable in the embedded feature space.

2.4.3. Feature learning via self-attention network

Recently, the SANet has played an important role in the field of multi-dimensional time series analysis [31][32], which is proposed by Ashish in Transformer [33]. In this paper, the SANet is selected as the feature extraction module of the Siamese network, and consists of two encoders and a fully connected output layer. And each encoder is composed of a multi-head self-attention mechanism, a Layer Norm, and a fully connected feed-forward network. In addition, the short-cut structure from the residual network is also added into the SANet [34]. The involved basic components are introduced as below.

The multi-head self-attention mechanism is the core component of the SANet, which allows the model to pay attention to information from different representation subspaces at different positions jointly [33]. The output $H_{k,i}$ of ith attention head is expressed by

$$H_{k,i} = Attention\left(Q_k W_i^Q, K_k W_i^K, V_k W_i^V\right) = softmax\left(\frac{Q_k W_i^Q \left(K_k W_i^K\right)^T}{\sqrt{d_Q}}\right) V_k W_i^V \quad (7)$$

where W_i^Q, W_i^K, W_i^V represent linear projections of ith attention head, and Q_k, K_k, V_k are Query, Key and Value of kth sample FSS_k, which obtained by linear projections of the $EFSS_k$. That is, $Q_k = (EFSS_k)\, W^Q$, $K_k = (EFSS_k)\, W^K$, $V_k = (EFSS_k)\, W^V$, and W^Q, W^K, W^V denote linear projections' parameter matrices with dimension $d_{model} \times d_Q$, $d_{model} \times d_K$, and $d_{model} \times d_V$. In addition, $EFSS_k$ with dimension $10 \times d_{model}$ is the sum of input embeddings of the sample FSS_k and positional encodings according to the literature [33].

The multi-head self-attention mechanism composed of h attention heads is expressed by

$$MH_k = MultiHead\left(Q_k, K_k, V_k\right) = Concat(H_{k,1}, H_{k,2}, \cdots, H_{k,h}) W^O \quad (8)$$

where W^O denotes the linear projection on concatenated all attention heads.

The Layer Norm is a normalization technique that allows all nodes in the same layer to share the same mean and variance. Experimental results in Ref.[35] showed that Layer Norm substantially reduces the training time. The output $\widetilde{MH}_k$ of Layer Norm is expressed by

$$\widetilde{MH}_k = \frac{\gamma \odot (MH_k - \mu)}{\sigma} + \beta \quad (9)$$

$$\mu = \frac{1}{N_h} \sum_{i=1}^{N_h} MH_k^{(i)} \quad (10)$$

$$\sigma = \sqrt{\frac{1}{N_h} \sum_{i=1}^{N_h} \left(MH_k^{(i)} - \mu\right)^2} \quad (11)$$

where $MH_k^{(i)}$ is the input of the ith hidden node of MH_k, and N_h represents the number of hidden nodes in this layer. In addition, β and γ are defined as the trainable bias and gain parameters, and $\odot$ is the element-wise multiplication between two vectors.

Fully connected feed-forward layer (FFL) contains two linear transformations. To improve the non-linear mapping and the learning ability of the module, a gaussian error linerar unit (GELU) activation function was added between the two linear transformations [36]. The calculation formula of this layer is expressed by

$$FFL(MH_k) = GELU((MH_k)\,W_1 + b_1)W_2 + b_2 \tag{12}$$

$$GELU((MH_k)\,W_1 + b_1) = [(MH_k)\,W_1 + b_1]\,\sigma^*\,(1.702 \times [(MH_k)\,W_1 + b_1]) \tag{13}$$

where W_1 and W_2 represent the weights for linear transformations, b_1 and b_2 represent biases, and $\sigma^*(\cdot)$ represents the cumulative distribution function of the logistic distribution.

2.5. The developed fault diagnosis approach

The proposed fault diagnosis framework not only integrates time series clustering and the DSSAN, but also adopts some tricks to improve the fault diagnosing performance on imbalanced datasets. First, only normal samples rather than fault samples are clustered in order to preserve more fault information as much as possible. Second, an early stopping strategy is adopted, which ends the training process once the training accuracy becomes 100%, so that the degree of overfitting can be further reduced. Finally, the model is evaluated using five-fold cross-validation to avoid the randomness of one experiment.

The overall procedure of the developed framework is depicted in Fig.4. The training process is divided into four steps: (1) **Clustering** The normal samples are clustered using the k-means, and the cluster centers are put together with the fault samples to construct sample pairs. (2) **Mapping** Feature extraction module maps sample pair $\left\langle FSS_k, \widehat{FSS}_k \right\rangle$ to an embedded feature space, and generates its embedded representation $\left\langle F_W\left(FSS_k\right), F_W\left(\widehat{FSS}_k\right)\right\rangle$. (3) **Similarity calculation** Similarity calculation module calculates the Euclidean distance between $F_W\left(FSS_k\right)$ and $F_W\left(\widehat{FSS}_k\right)$. (4) **Error back-propagation** The Siamese network is optimized by minimizing contrastive loss L_{sia}. Repeat steps 2 to 4 until the Siamese network converges.

In the test stage, the pth test sample $FSS_p^{(TEST)} \in D^{(TEST)}$ is paired with the samples in the support set $SD = CCD_{NOR} \bigcup D_{EIF}^{(TRAIN)} \bigcup D_{TSIF}^{(TRAIN)}$, and then calculate the Euclidean distances of the sample pairs. Then, it is necessary to find the test sample's most similar sample FSS_p^* in SD, where $FSS_p^* = \underset{FSS_k \in SD}{argmin}\left\{D_E\left(F_W(FSS_p^{(TEST)}), F_W(FSS_k)\right)\right\}$. Finally, $FSS_p^{(TEST)}$ is judged to belong to the same class with FSS_p^*.

3. Experimental results

3.1. Experimental data

In this paper, actual data is collected to evaluate the proposed fault diagnosis model, which comes from the real-life operation data of an Asia airline rather than simulated data. According to maintenance reports, a total of 1047 samples are constructed from the historical monitoring data, including 984 normal samples, 38 EGT indication fault samples, and 25 TAT sensor indication fault samples, which is summarized in Table 1. Then, the data set is divided into five disjoint subsets using five-fold cross-validation. Taking the first fold as an example, in the training set, the imbalance ratio between normal samples and EGT indication fault samples is 787/30≈26.2, while the imbalance ratio between normal samples and TAT Sensor indication fault samples is as high as 787/20≈39.3, so the training set is extremely unbalanced.

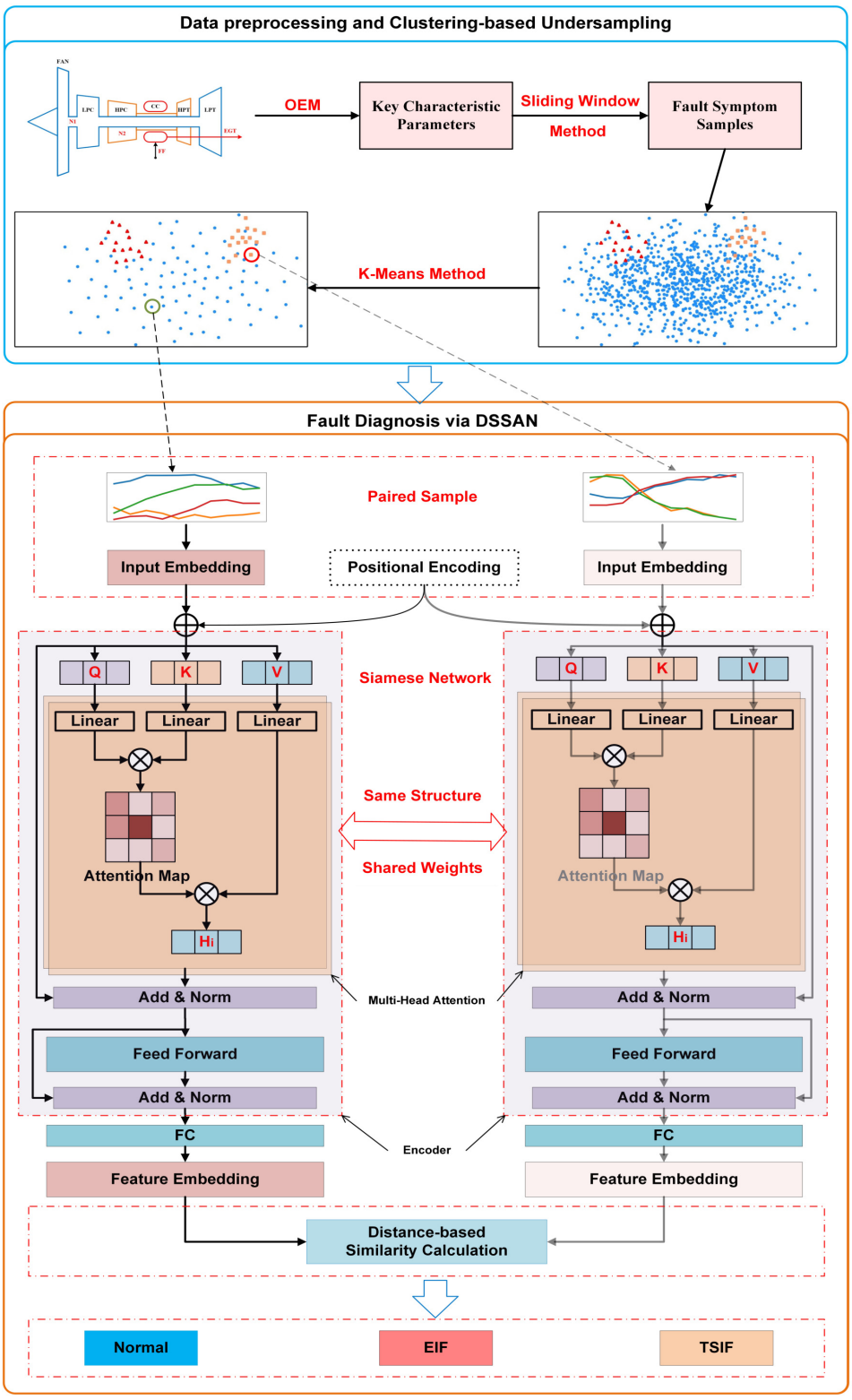


Figure 4. The procedures of the developed fault diagnosis approach

Table 1. Summary of the gas turbine dataset

Labels	Number of training samples					Number of test samples					Total
	K=1	K=2	K=3	K=4	K=5	K=1	K=2	K=3	K=4	K=5	
Normal	787	787	787	787	788	197	197	197	197	196	984
EIF	30	30	30	31	31	8	8	8	7	7	38
TSIF	20	20	20	20	20	5	5	5	5	5	25

After construction of sample pairs, there are 2200 sample pairs from same class, and 2100 sample pairs from different classes, which is summarized in Table 2. On one hand, there are 4300 sample pairs, and the sample size has been greatly expanded. On the other hand, the imbalance rate between same class sample pairs and different class sample pairs is only 2200/2100≈1.04, which effectively alleviates the class imbalance problem.

Table 2. Summary of the paired samples

Sample I	Sample II	Paired sample	Label	Total
Normal	Normal	<Normal, Normal >	1	30×30=900
Normal	EIF	<Normal, EIF>	0	30×30=900
Normal	TSIF	<Normal, TSIF>	0	30×20=600
EIF	EIF	<EIF, EIF>	1	30×30=900
EIF	TSIF	<EIF, TSIF>	0	30×20=600
TSIF	TSIF	<TSIF, TSIF>	1	20×20=400

3.2. Experimental setup

Experimental Platform Experimentation is implemented using Python 3.7 with PyTorch 1.8.0 and an Intel Core i7-9750H, 8G DDR4 processor.

Optimizer The neural network methods are optimized using Adam with an initial learning rate 0.0002, a mini-batch size of 128, and momentums β_1=0.5, β_2=0.999 [37]. And according to trial tests, we set the max-epoch 20 for the CBU-DSSAN.

Hyper-parameter setup In this paper, the grid search method is used to determine the hyper-parameters of the DSSAN. The DSSAN contains two encoders and a fully connected output layer. The multi-head self-attention mechanism in each encoder is composed of 4 attention heads, and the dimensionality of the input and output of multi-head self-attention mechanism is $d_{model} = 64$. In addition, for each of attention heads, $d_K = d_V = d_{model}/4 = 16$.

3.3. Evaluation metrics

For balanced datasets, overall accuracy can be used to evaluate the classification performance of fault diagnosis models. However, fault samples are far less than normal samples in the test set. In this case, overall accuracy is of no significant reference value. For example, the number of normal samples was 197, while the total number of fault samples was only 13. Even though all test samples were diagnosed as normal samples, overall accuracy was still as high as 93.81%. In this paper, the confusion matrix and some related metrics [38] are used to evaluate the proposed fault diagnosis model, which is summarized in Table 3.

In addition, true normal sample rate (*TNR*), true EIF sample rate (*TER*), and true TSIF sample rate (*TTR*) are used to evaluate the diagnosis effectiveness of normal samples, EGT indication fault samples, and TAT sensor indication fault samples, respectively. The overall diagnosis effect of the model can be evaluated by the average of the three, which is recorded as the balanced accuracy. These metrics can be simply calculated according to the confusion matrix, and the specific calculation formula is expressed by

Table 3. An illustration of confusion matrix

Actual label	Predicted label		
	Normal	EIF	TSIF
Normal	N_{11}	N_{12}	N_{13}
EIF	N_{21}	N_{22}	N_{23}
TSIF	N_{31}	N_{32}	N_{33}

$$TNR = \frac{N_{11}}{N_{11} + N_{12} + N_{13}} \tag{14}$$

$$TER = \frac{N_{22}}{N_{21} + N_{22} + N_{23}} \tag{15}$$

$$TTR = \frac{N_{33}}{N_{31} + N_{32} + N_{33}} \tag{16}$$

$$balanced\ accuracy = \frac{TNR + TER + TTR}{3} \tag{17}$$

where N_{ij} is the number of samples, which actually belong to the ith class and are classified into the jth class.

3.4. Experimental comparisons

To verify the effectiveness of the proposed CBU-DSSAN model, two groups of comparative experiments are conducted in this paper.

(1) To validate the necessity of balancing different classes, unbalanced data are directly used to train deep neural networks, including ConvNet, RNN, LSTM, GRU, and SANet.

(2) To show the superiority of the developed method over classical class balancing methods, SMOTE-based over-sampling and ClusterCentroids-based downsampling are combined with SANet without Siamese architecture, which are separately denoted as OS-SANet and DS-SANet, respectively. In addition, they are implemented by the SMOTE and ClusterCentroids in imblearn package. The number of samples in all three classes is 787 after over-sampling, while the number of samples in the three classes is 20 after down-sampling.

Comparative experimental results are shown in Fig.5-9 and Table 4. First, Fig. 5-8 show experimental results in terms of *TNR*, *TER*, *TTR*, and confusion matrices. Second, the balanced accuracy of the different fault diagnosis methods is listed in Table 4. Finally, the distributions of samples in the feature spaces have been depicted in Fig. 9.

For EGT indication fault samples, the proposed method outperforms the first six methods in most cases; compared with DS-SANet, it achieves better results in the fifth experiment. For TAT sensor indication fault samples, our method and DS-SANet have the same best performance, and outperform any of the first six methods at least for one time. For normal samples, the proposed method achieves better diagnosis performance than the DS-SANet but is not as good as the first six methods. Because the first six methods pay more attention to the training of normal samples, which causes fault samples are easily misclassified as normal samples. The safety of gas turbines is extremely important, and the first six methods are not suitable for gas turbine fault diagnosis. That is, to reduce the missed fault samples, we would rather have some misjudged normal samples but not vice versa.

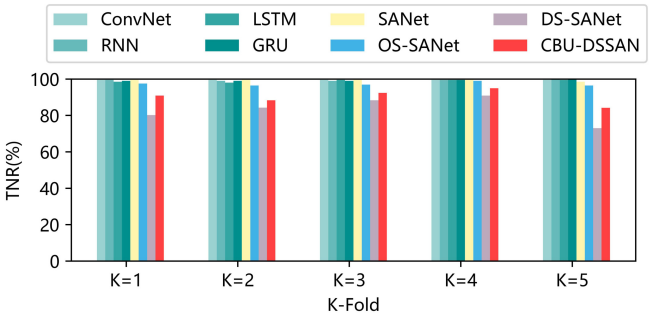


Figure 5. TNR of these methods in five-fold cross-validation

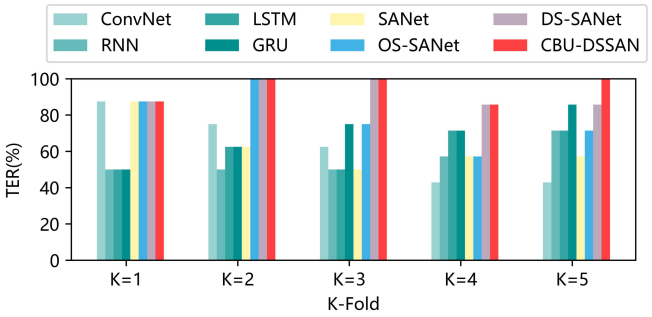


Figure 6. TER of these methods in five-fold cross-validation

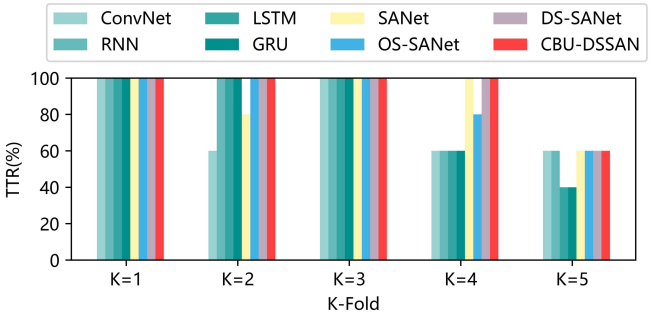


Figure 7. TTR of these methods in five-fold cross-validation

Table 4 shows the balanced accuracy of different fault diagnosis methods, which can be concluded as follows. First, compared with the ConvNet, RNN, LSTM, and GRU, SANet achieves the best results in terms of balanced accuracy, improving by 4.17%, 3.68%, 3.36%, and 0.64%, respectively. Obviously, SANet benefits from the multi-head self-attention mechanism. In other words, the multi-head self-attention mechanism can capture more time-series correlations and dependencies from multi-dimensional time series, which is very suitable for actual monitoring data. Second, compared with SANet, the diagnosis performance of OS-SANet and DS-SANet has been improved by 4.44% and 6.61% respectively. This illustrates the importance of balancing the data set; that is, balancing fault samples and normal samples improves the diagnosis accuracy in this experiment. Third, the diagnostic performance of the DS-SANet is better than that of the OS-SANet in this experiment, which shows the usefulness of clustering-based downsampling. Finally, the CBU-DSSAN achieves the best results in terms of balanced accuracy. Compared with all

the considered methods, the performance of CBU-DSSAN is increased by 13.05%, 12.56%, 12.25%, 9.53%, 8.88%, 4.44%, and 3.22%, respectively. This shows that the proposed method is helpful to resolve the problem of the multi-class classification under class-imbalanced conditions.

Table 4. Balanced accuracy and inference time of the different fault diagnosis methods

Method	balanced accuracy(%)						Inference Time
	K=1	K=2	K=3	K=4	K=5	Average	(milliseconds)
ConvNet	**95.66**	78.16	87.33	67.44	67.44	79.21	**0.02**
RNN	83.16	82.99	82.99	72.21	77.14	79.70	0.07
LSTM	82.82	86.82	83.16	76.97	70.30	80.01	0.34
GRU	82.99	87.16	91.32	76.97	75.23	82.73	0.15
SANet	**95.66**	80.66	83.16	85.54	71.87	83.38	0.13
OS-SANet	94.98	**98.81**	90.65	78.70	75.95	87.82	0.06
DS-SANet	89.23	94.75	96.10	92.19	72.89	89.04	0.07
CBU-DSSAN	92.78	96.10	**97.46**	**93.54**	**81.39**	**92.26**	1.33

Table 4 also shows the inference time of each fault diagnosis model in the test set. It is easy to observe that the proposed method takes about 1.33 milliseconds to detect one test sample. Although the proposed method is 1.31 milliseconds slower than the ConvNet, its speed is still very fast, which can be acceptable in engineering practice.

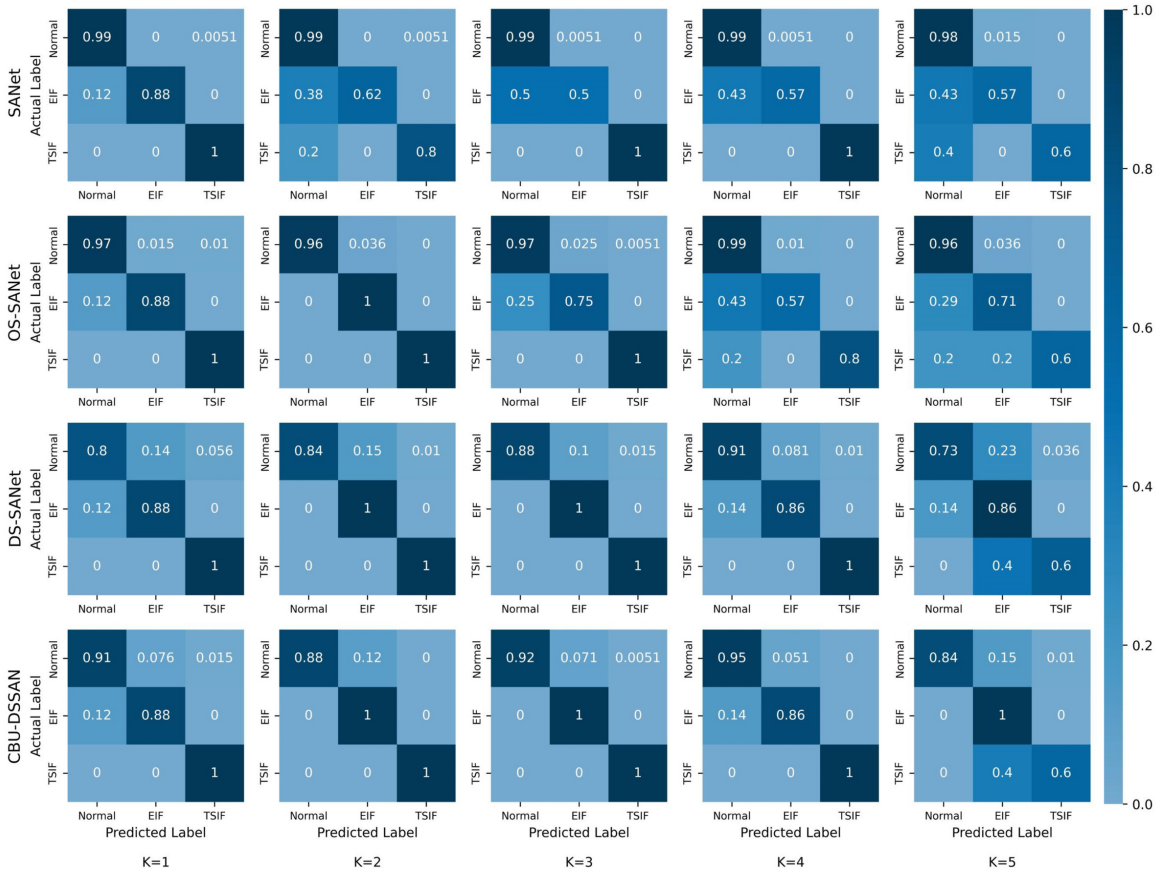


Figure 8. Confusion matrices of the considered methods on a testing dataset

To more intuitively observe the fault diagnosis effect of each model, Fig.8 shows the confusion matrices of SANet, OS-SANet, DS-SANet, and CBU-DSSAN in the tests. It can be seen that the performances of the CBU-DSSAN are mostly better than that of the SANet, OS-SANet, and DS-SANet for fault samples. Taking $K = 5$ as an example, the *TER* of the four models are 57%, 71%, 86% and 100% respectively, and their *TTR* are 60%, 60%, 60%, and 60%, respectively. Therefore, there is still room for improvement in fault diagnosis of gas turbines.

Although the two-dimensional visualizations using t-distributed stochastic neighbor embedding may lose some information, they can provide a perspective on the distributions of samples in the high-dimensional spaces. As shown in Fig.9, fault samples and normal samples are highly overlapped in the SANet, OS-SANet, and DS-SANet. In this case, fault samples can easily be misclassified as normal samples. In contrast, for the CBU-DSSAN, fault samples and the normal samples have little overlap and are easier to be correctly classified.

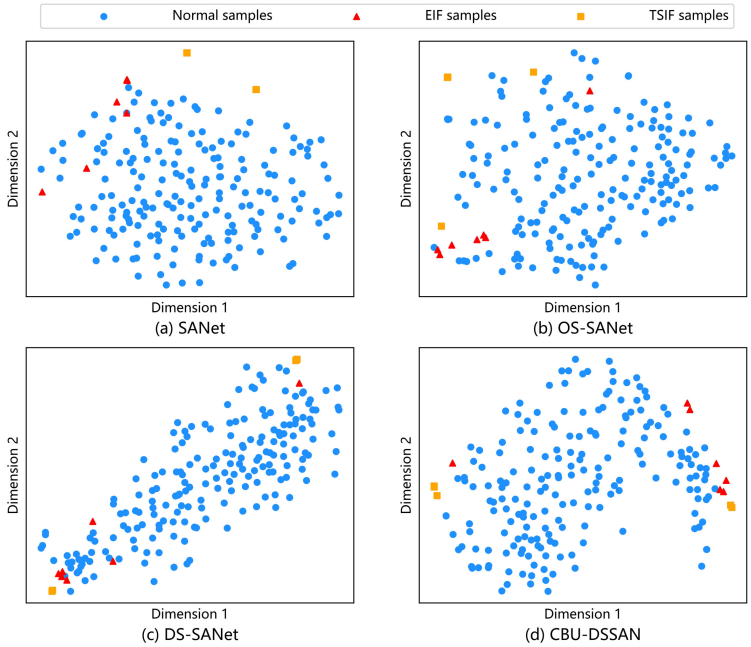


Figure 9. Distributions of the fifth-fold testing samples in the two-dimensional feature spaces

3.5. Investigation on the performance with different numbers of cluster centers

The number of cluster centers is an important hyper-parameter and should be chosen with considerations from at least two aspects. First, the number of cluster centers should not be too large; otherwise, it will not balance the data set. Second, the number of cluster centers should not be too small; otherwise, it will lose too much important information of normal samples. Therefore, the influence of numbers of cluster centers on the diagnostic performance is investigated in this section.

As summarized in Table 5, when the number of cluster centers is set to 30, the CBU-DSSAN achieves the best balanced accuracy. These cluster centers not only balance the data set but also retains important information. It is notable that the diagnosis performance is not only influenced by the number of cluster centers but also closely related to the splitting condition of the data set. Here, the reason is that each faulty class only has a very small number of training samples. It is unavoidable that the fault sample distribution of the training data set sometimes deviates from the overall distribution, which will lead to relatively poor performances.

Table 5. A summary of average results using different numbers of cluster centers

Number of cluster centers	Balanced accuracy(%) K=1	K=2	K=3	K=4	K=5	Average
20	**97.80**	94.41	95.60	92.70	74.25	90.95
25	95.09	94.92	96.10	91.68	75.78	90.71
30	92.78	96.10	**97.46**	93.54	**81.39**	**92.26**
35	92.95	95.60	96.78	88.95	73.23	89.51
40	93.80	**96.61**	89.97	**94.39**	72.89	89.54
45	93.29	95.93	93.29	**94.39**	78.33	91.05
50	94.31	93.80	97.12	89.29	78.16	90.54

4. Conclusions

This paper develops a novel imbalanced fault diagnosis framework for gas turbines that integrates the clustering-based downsampling and deep Siamese self-attention network. First, after downsampling, the Siamese network is trained on a balanced training dataset. Second, the well-trained Siamese network is used to measure the similarity between the test sample and the sample in the support set, and the test sample is judged to belong to the class of its most similar sample. Third, the five-fold cross-validation is carried out to verify the effectiveness of the proposed CBU-DSSAN.

In the experiment, the CBU-DSSAN achieved improvements of 8.88%, 4.44%, and 3.22% compared with the general SANet, OS-SANet, and DS-SANet in terms of average balanced accuracy. One reason is that the clustering-based downsampling is used to find a set of cluster centers of normal training samples, and these cluster centers are put together with the fault samples to serve as a balanced training dataset. The other reason is that the deep Siamese self-attention network maps the original data set into an embedded feature space, in which the separability of samples from different classes is enhanced.

Finally, fault diagnosis of gas turbines is a long-standing challenging task. To further improve the diagnosing accuracy, our future efforts will concentrate on the discovery of a more effective method that can simultaneously optimize the hyperparameters of clustering-based downsampling and deep Siamese self-attention network.

References

[1] Q. Yang, S. Li, Y. Cao, A strong tracking filter based multiple model approach for gas turbine fault diagnosis, Journal of Mechanical Science and Technology 32 (1) (2018) 465–479.
[2] M. Montazeri-Gh, S. Yazdani, Application of interval type-2 fuzzy logic systems to gas turbine fault diagnosis, Applied Soft Computing 96 (2020) 106703.
[3] A. D. Fentaye, A. T. Baheta, S. I. Gilani, K. G. Kyprianidis, A review on gas turbine gas-path diagnostics: State-of-the-art methods, challenges and opportunities, Aerospace 6 (7) (2019) 83.
[4] H. Zhou, Y. Ying, J. Li, Y. Jin, Long-short term memory and gas path analysis based gas turbine fault diagnosis and prognosis, Advances in Mechanical Engineering 13 (8) (2021) 16878140211037767.
[5] D. L. Simon, A. W. Rinehart, A model-based anomaly detection approach for analyzing streaming aircraft engine measurement data, Vol. 45752, American Society of Mechanical Engineers, 2014.
[6] Z. Tan, S. Zhong, L. Lin, Trans-layer model learning: A hierarchical modeling strategy for real-time reliability evaluation of complex systems, Reliability Engineering & System Safety 182 (2019) 120–132.
[7] Z. Tian, T. Jin, B. Wu, F. Ding, Condition based maintenance optimization for wind power generation systems under continuous monitoring, Renewable Energy 36 (5) (2011) 1502–1509.
[8] X. Li, Q. Ding, J.-Q. Sun, Remaining useful life estimation in prognostics using deep convolution neural networks, Reliability Engineering & System Safety 172 (2018) 1–11.
[9] Y. LeCun, Y. Bengio, G. Hinton, Deep learning, nature 521 (7553) (2015) 436–444.
[10] L. Meng, M. Zhao, Z. Cui, X. Zhang, S. Zhong, Empirical mode reconstruction: Preserving intrinsic components in data augmentation for intelligent fault diagnosis of civil aviation hydraulic pumps, Computers in Industry 134 (2022) 103557.
[11] K. Zhang, B. Tang, L. Deng, X. Liu, A hybrid attention improved resnet based fault diagnosis method of wind turbines gearbox, Measurement 179 (2021) 109491.
[12] M. Zhao, S. Zhong, X. Fu, B. Tang, M. Pecht, Deep residual shrinkage networks for fault diagnosis, IEEE Transactions on Industrial Informatics 16 (7) (2019) 4681–4690.

[13] K. Zhang, B. Tang, L. Deng, Q. Tan, H. Yu, A fault diagnosis method for wind turbines gearbox based on adaptive loss weighted meta-resnet under noisy labels, Mechanical Systems and Signal Processing 161 (2021) 107963.
[14] R. Zhao, D. Wang, R. Yan, K. Mao, F. Shen, J. Wang, Machine health monitoring using local feature-based gated recurrent unit networks, IEEE Transactions on Industrial Electronics 65 (2) (2017) 1539–1548.
[15] H. Yu, Z. Tian, H. Li, B. Xu, G. An, A novel deep belief network model constructed by improved conditional rbms and its application in rul prediction for hydraulic pumps., International Journal of Acoustics and Vibration 25 (3) (2020) 373–383.
[16] Z. Li, S. S. Zhong, L. Lin, Novel gas turbine fault diagnosis method based on performance deviation model, Journal of Propulsion and Power (2016) 1–10.
[17] Wei-Chao, Lin, Chih-Fong, Tsai, Ya-Han, Hu, Jing-Shang, Jhang, Clustering-based undersampling in class-imbalanced data - sciencedirect, Information Sciences 409-410 (2017) 17–26.
[18] Y. Djenouri, A. Belhadi, P. Fournier-Viger, J. C.-W. Lin, Fast and effective cluster-based information retrieval using frequent closed itemsets, Information Sciences 453 (2018) 154–167.
[19] X. Liu, Y. Zhou, J. Zhao, R. Yao, B. Liu, Y. Zheng, Siamese convolutional neural networks for remote sensing scene classification, IEEE Geoscience and Remote Sensing Letters 16 (8) (2019) 1200–1204.
[20] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, W. Zhang, Informer: Beyond efficient transformer for long sequence time-series forecasting, in: Proceedings of Association for the Advancement of Artificial Intelligence, 2021.
[21] F. Thabtah, S. Hammoud, F. Kamalov, A. Gonsalves, Data imbalance in classification: Experimental evaluation, Information Sciences 513 (2020) 429–441.
[22] M. M. Bejani, M. Ghatee, A systematic review on overfitting control in shallow and deep neural networks, Artificial Intelligence Review 54 (8) (2021) 6391–6438.
[23] Y. Liu, C. Gong, L. Yang, Y. Chen, Dstp-rnn: A dual-stage two-phase attention-based recurrent neural network for long-term and multivariate time series prediction, Expert Systems with Applications 143 (2020) 113082.
[24] S. Zhong, D. Liu, L. Lin, M. Zhao, X. Fu, F. Guo, A novel anomaly detection method for gas turbines using weight agnostic neural network search, Asia-Pacific International Symposium on Advanced Reliability and Maintenance Modeling, IEEE, 2020, pp. 1–6.
[25] S. Zhong, D. Liu, L. Lin, M. Zhao, X. Fu, F. Guo, Cae-wann: A novel anomaly detection method for gas turbines via search space extension, Quality and Reliability Engineering International.
[26] J. Paparrizos, L. Gravano, k-shape: Efficient and accurate clustering of time series, in: Proceedings of the 2015 ACM SIGMOD international conference on management of data, 2015, pp. 1855–1870.
[27] G. Liu, J. Yang, Y. Hao, Y. Zhang, Big data-informed energy efficiency assessment of china industry sectors based on k-means clustering, Journal of cleaner production 183 (2018) 304–314.
[28] P. M. Shakeel, S. Baskar, V. Dhulipala, M. M. Jaber, Cloud based framework for diagnosis of diabetes mellitus using k-means clustering, Health information science and systems 6 (1) (2018) 1–7.
[29] X. Zhou, W. Liang, S. Shimizu, J. Ma, Q. Jin, Siamese neural network based few-shot learning for anomaly detection in industrial cyber-physical systems, IEEE Transactions on Industrial Informatics 17 (8) (2020) 5790–5798.
[30] R. Hadsell, S. Chopra, Y. LeCun, Dimensionality reduction by learning an invariant mapping, Vol. 2 of IEEE Computer Society Conference on Computer Vision and Pattern Recognition, IEEE, 2006, pp. 1735–1742.
[31] S. Li, X. Jin, Y. Xuan, X. Zhou, W. Chen, Y.-X. Wang, X. Yan, Enhancing the locality and breaking the memory bottleneck of transformer on time series forecasting, Advances in Neural Information Processing Systems 32 (2019).
[32] J. Xu, J. Wang, M. Long, et al., Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting, Advances in Neural Information Processing Systems 34 (2021).
[33] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, Advances in neural information processing systems 30 (2017).
[34] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016.
[35] J. L. Ba, J. R. Kiros, G. E. Hinton, Layer normalization, arXiv preprint arXiv:1607.06450 (2016).
[36] D. Hendrycks, K. Gimpel, Gaussian error linear units (gelus), arXiv preprint arXiv:1606.08415 (2016).
[37] S. Akcay, A. Atapour-Abarghouei, T. P. Breckon, Ganomaly: Semi-supervised anomaly detection via adversarial training, in: Asian conference on computer vision, Springer, 2018, pp. 622–637.
[38] M. Grandini, E. Bagli, G. Visani, Metrics for multi-class classification: an overview, arXiv preprint arXiv:2008.05756 (2020).